1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

Started on	Friday, 25 September 2026, 8:16 AM
State	Finished
Completed on	Friday, 25 September 2026, 8:22 AM
Time taken	5 mins 37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	11 10 9 7 6 5 1 2 3 8 4 7	7	7	
	5 1 2 3 4 4	4	4	
	5 1 1 2 3 4	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

Started on	Friday, 25 September 2026, 8:22 AM

State	Finished
Completed on	Friday, 25 September 2026, 8:23 AM
Time taken	1 min 4 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	11 10 9 7 6 5 1 2 3 8 4 7	7	7	
	5 1 2 3 4 4	4	4	
	5 1 1 2 3 4	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

Started on	Friday, 25 September 2026, 8:24 AM
State	Finished
Completed on	Friday, 25 September 2026, 9:25 AM
Time taken	1 hour

Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	
	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

Started on	Friday, 25 September 2026, 8:32 AM
State	Finished
Completed on	Friday, 25 September 2026, 9:24 AM
Time taken	52 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	
	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

Started on	Friday, 25 September 2026, 9:28 AM
State	Finished
Completed on	Friday, 25 September 2026, 9:29 AM
Time taken	51 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	3 1 3 5 4	1	1	
	10 1 4 6 8 12 14 15 20 21 25 1	1	1	

	Input	Expected	Got	
	10 1 2 3 5 11 14 16 24 28 29 0	0	0	
	10 0 2 3 7 13 14 15 20 24 25 10	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Started on	Friday, 25 September 2026, 8:40 AM
State	Finished
Completed on	Friday, 25 September 2026, 9:03 AM
Time taken	22 mins 49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	3 1 3 5 4	1	1	
	10 1 4 6 8 12 14 15 20 21 25 1	1	1	
	10 1 2 3 5 11 14 16 24 28 29	0	0	

	Input	Expected	Got	
	0			
	10 0 2 3 7 13 14 15 20 24 25 10	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

cc